## Object Class

- java.lang (System)
- Parent → Object
- Every class in java inherits from Object class.

```
class Student extends Object {
    =
}
```

Integer System → Object

```
class Student extend Human {

}
```

## Object

- Common behaviours (Inheritance)
- Object obj = new Student();
  (polymorphism)

```
class Object {
    =
}
```

### Core Methods

- toString()
- equals()
- hashCode()
- getClass()

### Cloning

- clone()

### Garbage Collection

- finalize()

### Threads

- wait()
- notify()
- notifyAll()

## toString()

↳ Convert anything to String.

```
public String toString() {
    =
}
```

↳ String Representation of any Object

```
class Object {
    public String toString() {
        // String Representation of Object.
    }
}

class Student extend Object {
    @override
    toString() {
        =
    }
}
```

```
Student s1 = new Student();
s1.toString();
```

⇓

ClassName@Hex Code

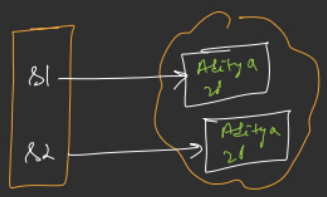
## equals()

↳ Compare 2 objects & returns true / false.

```
class Object {
    public boolean equals (Object obj) {
        //Default
        }
}
```

↳ Compare references.

```
class Student {
    String name;
    int age;
}
```




```
Student s1 = new Student ("Aditya", 28);
Student s2 = new Student ("Aditya", 28);

s1.equals(s2);
```

$\boxed{s1 == s2;}$ // Reference

## hashCode()

↳ Return an integer of an Object

↳ HexaDecimal format

```
class Object {
    public int hashCode() {
        // default.
    }
}
```

Rule :

↳ If 2 objects are equal then their hashCode must be equal.

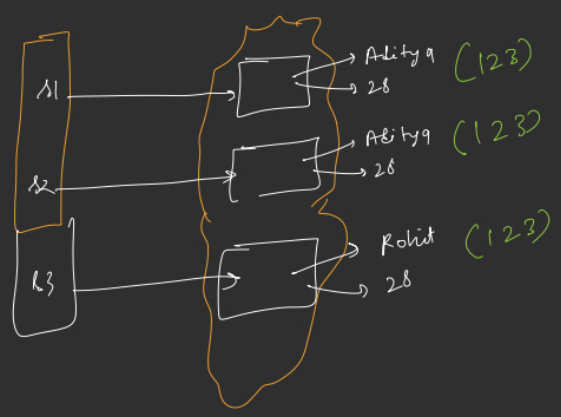
```
Student s1 = new Student (Aditya, 28);
Student s2 = new Student (Aditya, 28);
```

s1.equals(s2) → true

s1.hashCode() == s2.hashCode()

HashSet
Hashmap
Hashtable

↳ But the reverse might not be true.




## getClass()

↳ Return runtime class of an Object.

```
Student s1 = new Student();
s1.getClass().getName()
                    // Student
```

```
class Object {
    public final Class<?> getClass() {
        // default
    }
}
```

## Clone()

↳ create copy of an Object. → Shallow Copy

```
class Object {
    protected Object clone() throws CloneNotSupportedException {
        // default → if( this obj implement Clonable) allow it
    }
}
```

else throw Exception

**Clonable (Interface)**

```
class Student implement Cloneable {
    Clone() {
    }
}
```

**Not every Object should be Cloneable**

↳ Database
↳ Thread

```
Class Student implements Cloneable {
    // Override → (Deep Copy)
}
```

```
interface Cloneable {
    // empty
}
```

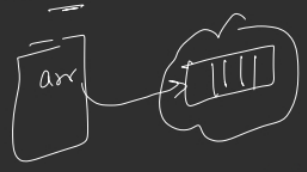
→ Marker Interfaces

## Array → Object

Arrays

↳ Non-primitive.

```
int[] arr = new int[5];
```




int, char, float } NO → Integer, Character, Float

## finalize()

↳ Garbage Collection

Deprecated

↳ unpredictible, unsafe, unreliable

wait()
notify()
notifyAll() → Threads